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Foreword

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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

3GPP acknowledges the contribution of the Parlay X Web Services specifications from The Parlay Group. The Parlay Group is pleased to see 3GPP acknowledge and publish the present document, and the Parlay Group looks forward to working with the 3GPP community to improve future versions of the present document.

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

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Introduction

The present document is part 14 of a multi-part deliverable covering the 3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; Open Service Access (OSA); Parlay X Web Services, as identified below:

Part 1: "Common"
Part 2: "Third party call"
Part 3: "Call Notification"
Part 4: "Short Messaging"
Part 5: "Multimedia Messaging"
Part 6: "Payment"
Part 7: "Account management"
Part 8: "Terminal Status"
Part 9: "Terminal location"
Part 10: "Call handling"
Part 11: "Audio call"
Part 12: "Multimedia conference"
Part 13: "Address list management"
Part 14: "Presence"
Part 15: "Message Broadcast"
Part 16: "Geocoding"
Part 17: "Application driven Quality of Service (QoS)"
Part 18: "Device Capabilities and Configuration"
Part 19: "Multimedia streaming control"
Part 20: "Multimedia multicast session management"
Part 21: "Content management"
Part 22: "Policy"

1 Scope

The present document is Part 14 of the Stage 3 Parlay X Web Services specification for Open Service Access (OSA).

The OSA specifications define an architecture that enables application developers to make use of network functionality through an open standardized interface, i.e. the OSA APIs. The concepts and the functional architecture for the OSA are contained in 3GPP TS 23.198 [3]. The requirements for OSA are contained in 3GPP TS 22.127 [2].

The present document specifies the Presence Web Service aspects of the interface. All aspects of the Presence Web Service are defined here, these being:

- Name spaces.
- Sequence diagrams.
- Data definitions.
- Interface specification plus detailed method descriptions.
- Fault definitions.
- Service policies.
- WSDL Description of the interfaces.

The present document has been defined jointly between 3GPP TSG CT WG5, ETSI TISPAN and the Parlay Consortium.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 22.127: "Service Requirement for the Open Services Access (OSA); Stage 1".

[3] 3GPP TS 23.198: "Open Service Access (OSA); Stage 2".

[4] 3GPP TS 22.101: "Service aspects; Service principles".

[5] W3C Recommendation (2 May 2001): "XML Schema Part 2: Datatypes". http://www.w3.org/TR/2001/REC-xmlschema-2-20010502/.

[6] 3GPP TS 29.199-1: "Open Service Access (OSA); Parlay X Web Services; Part 1: Common".

[7] Void.

[8] 3GPP TS 29.198-14: "Open Service Access (OSA) Application Programming Interface (API); Part 14: Presence and Availability Management (PAM)".

[9] RFC 3856: "A Presence Event Package for the Session Initiation Protocol (SIP)". http://www.ietf.org/rfc/rfc3856.txt

[10] Void.

[11] Void.

[12] 3GPP TS 23.141: "Presence service; Architecture and functional description; Stage 2".

[13] 3GPP TS 29.199-13: "Open Service Access (OSA); Parlay X Web Services; Part 13: Address list management".

[14] IETF RFC 3265: "Session Initiation Protocol (SIP)-Specific Event Notification".

[15] Void.

[16] IETF RFC 3863: "Presence Information Data Format (PIDF) ".

[17] IETF RFC 4480: "RPID: Rich Presence Extensions to the Presence Information Data Format (PIDF)".

[18] OMA-SUP-XSD_prs_presrules-V1_0: "Presence SIMPLE – Presrules", Version 1.0, Open Mobile Alliance™. http://www.openmobilealliance.org/tech/profiles/prs_presrules-v1_0.xsd".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TS 29.199-1 [6] and the following apply:

applications: for Instant Messaging, Push to Talk, or call control and other purposes may become clients of the presence Web Service
We assume that these applications belong to a watcher and authenticate to the services in the name of the watcher.

identity: represents a user in the real world

NOTE: See OSA/Parlay PAM identities [8], section 4.4.1.

presence attributes: contain information about a presentity
An attribute has a name and a value and can be supplied by any device, application or network module that can be associated to the presentity's identity. A watcher can obtains attributes only after he has successfully subscribed to them. Examples for attributes are activity, location type, communication means, etc.

presence information: consists of a set of attributes that characterize the presentity such as current activity, environment, communication means and contact addresses
Only the system and the presentity have direct access to this information, which may be collected and aggregated from **several** devices associated to the presentity.

subscription: before a watcher can access presence data, he has to subscribe to it
One possibility the API provides is an end-to-end subscription concept, in which only identities that have accepted a subscription to their presence can be addressed. Subscriptions can be also automatically handled by server policies edited by the presentity or other authorized users. The service/protocol to manage those policies is out of the scope of the present document.

NOTE: This definition is not related to the term "subscription" in 3GPP TR 21.905 [1].

watcher and presentity: We use these names to denote the role of the client connected to the presence services
Like in OSA/Parlay PAM [8] the watcher and the presentity have to be associated to identities registered to the system, i.e. users, groups of users or organizations.

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TS 29.199-1 [6] and the following apply:

ACL	Access Control List
DMS	Data Manipulation Server
GM	Group Management
IETF	Internet Engineering Task Force
IMS	IP Multimedia Subsystem
ISC	IP multimedia subsystem Service Control interface
MMS	Multimedia Message Service
PAM	Presence and Availability Management
RLS	Resource List Server
SCF	Service Capability Feature
SIMPLE	SIP for Instant Messaging and Presence Leveraging Extensions
SIP	Session Initiation Protocol
SMS	Short Message Service
URI	Uniform Resource Identifier
WS	Web Service
WSDL	Web Services Definition Language
XCAP	XML Configuration Access Protocol
XML	eXtensible Markup Language
XMPP	eXtensible Messaging and Presence Protocol
XSD	XML Schema Definition

4 Detailed service description

The presence service allows for presence information to be obtained about one or more users and to register presence for the same. It is assumed that the typical client of these interfaces is either a supplier or a consumer of the presence information. An Instant Messaging application is a canonical example of such a client of this interface.

Figure 4.1 shows the architecture of the presence Web Service and the underlying services. The OSA/Parlay PAM SCF is the straightforward option and implements the presence server with extended identity, device capability, and presence agent management. OSA/Parlay PAM allows aggregation of presence information from internet, mobile and enterprise users, etc. using a presence transport network of SIP or XMPP servers. The Presence Web Service can however communicate directly for example with IMS presence network elements (presence and resource list servers) using the ISC (SIP/SIMPLE) protocol interface.

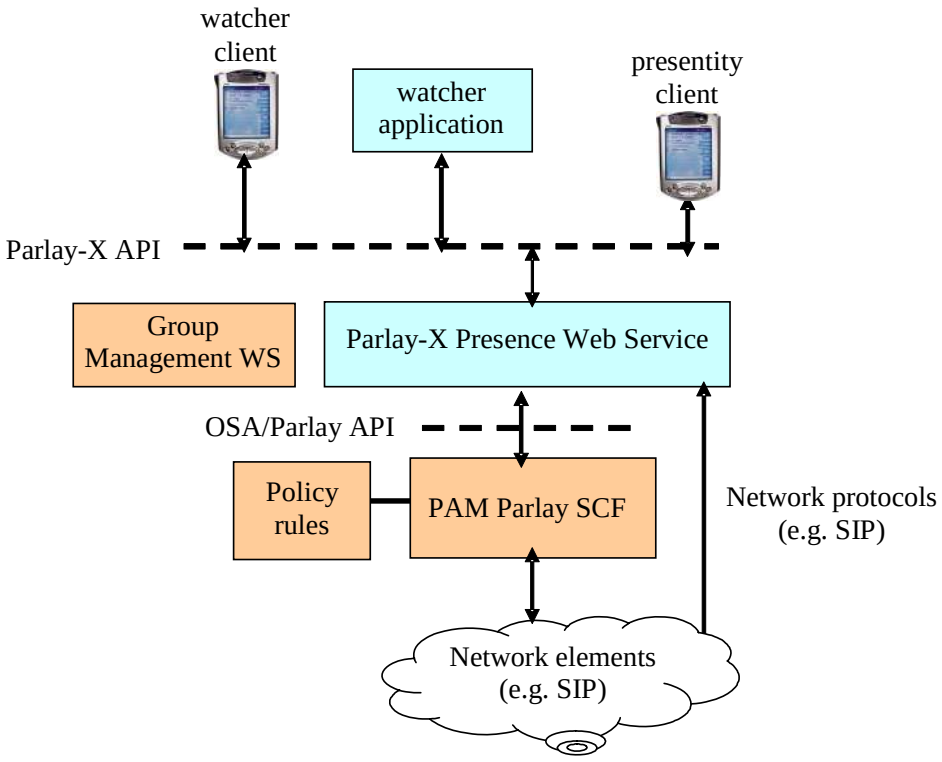


Figure 4.1: The PAM Web Service Environment

Relationship to Similar or Supplanted Specifications:

The most important relations are to:

- Parlay-X Terminal Status and Terminal Location: Both services deal with information that could be considered part of the user's presence information. Communication abilities can be derived from terminal status information, and the user's placetype can be derived from his location.
- OSA/Parlay PAM: The OSA/Parlay Presence and Availability specification can be considered the big brother of this specification. While ParlayX Presence stays behind OSA PAM in terms of flexibility and power - especially concerning attributes and management interfaces - it also extends PAM by introducing end-to-end authorization. This specification aims to be mappable to OSA PAM.
- SIP SIMPLE [9]: This specification aims to be mappable to the SIP/SIMPLE architecture.
- XMPP (Jabber): Many principles of this specification (see Bibliography) have been adopted, especially the end-to-end authorization.
- IETF Rich Presence (see Bibliography). The set of attributes the present document specifies is closely aligned with the IETF's Rich Presence ideas.
- Group Management [13]: Presence of groups is supported by this specification, however their creation and manipulation has to be done using the GM PX Web Service. In the 3GPP presence context, contact lists and group manipulation is done with the XCAP protocol (see Bibliography).

5 Namespaces

The PresenceConsumer interface uses the namespace:

http://www.csapi.org/wsdl/parlayx/presence/consumer/v3_2

The PresenceNotification interfaces use the namespace:

http://www.csapi.org/wsdl/parlayx/presence/notification/v3_2

The PresenceSupplier interfaces use the namespace:

http://www.csapi.org/wsdl/parlayx/presence/supplier/v3_2

The PresenceSupplierNotificationManager interfaces use the namespace:

http://www.csapi.org/wsdl/parlayx/presence_supplier/notification_manager/v3_0

The PresenceSupplierNotification interfaces use the namespace:

http://www.csapi.org/wsdl/parlayx/presence_supplier/notification/v3_0

The data types are defined in the namespace:

http://www.csapi.org/schema/parlayx/presence/v3_2

The 'xsd' namespace is used in the present document to refer to the XML Schema data types defined in XML Schema [5]. The use of the name 'xsd' is not semantically significant.

6 Sequence diagrams

6.1 Interface flow overview

The sequence diagram shows the interactions in case both watcher application and presentity are Web Service clients. Compared to the SIP interactions, the subscription notification is separated from the delivery of presence information itself. Based on the subscription result, the watcher can select the polling or notification mode for presence events. Changes in the authorization of presence attributes are propagated to the watchers via updateAuthorisationRule() message, the blocking of a subscription by the presentity are propagated via an endSubscriptionNotification message.

The sequence diagram does not show the internal communication within the presence server. It is assumed that the Presence Consumer and Supplier interfaces are implemented by the same instance. If an implementer of the API finds other solutions preferable, he has to take care of the internal communication himself.

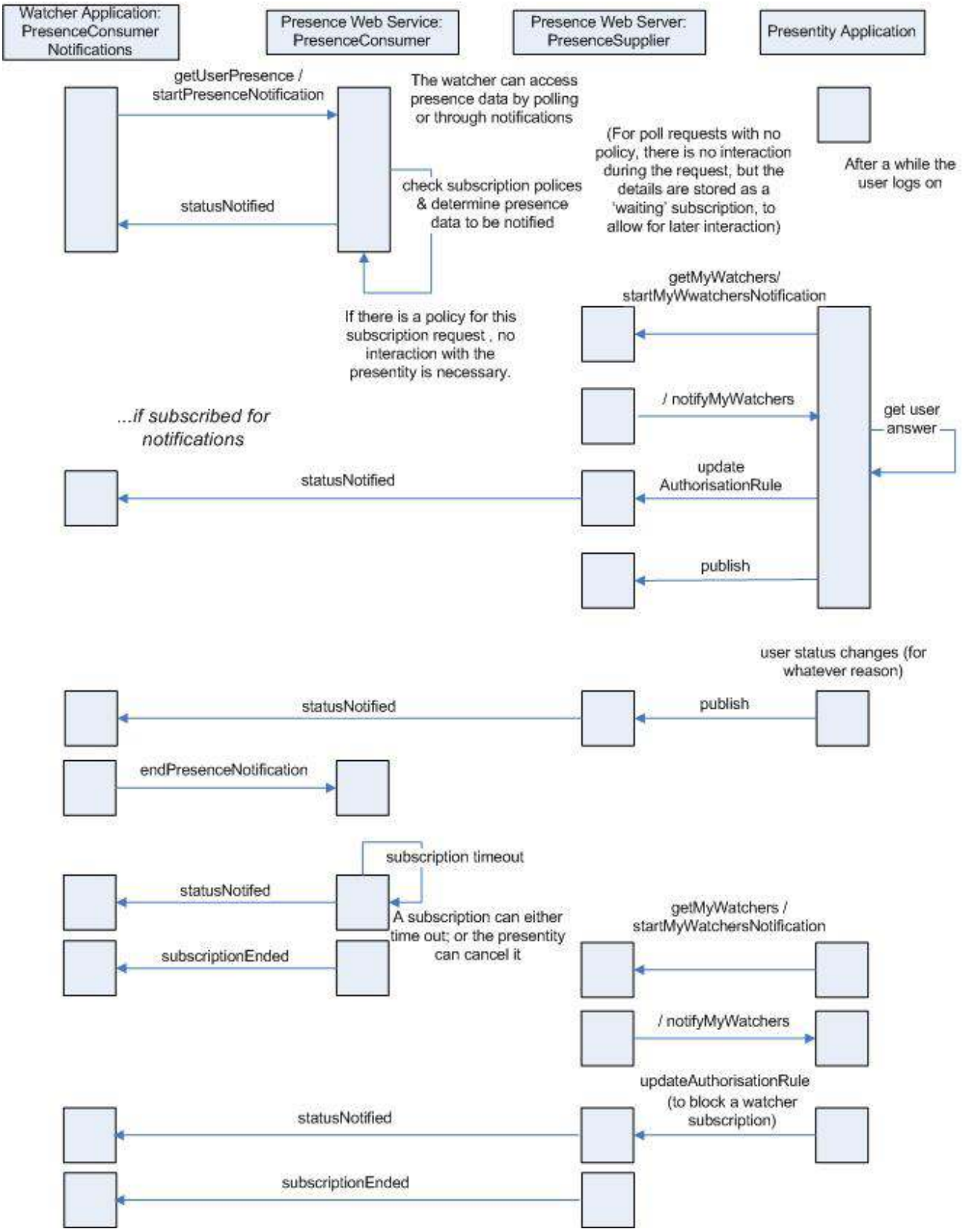


Figure : Message interaction overview

7 XML Schema data type definition

Presence attributes are inspired by the IETF's Rich Presence ideas (see Bibliography).

7.1 PresenceAttributeType enumeration

The different types of attributes. For each entry in this enumeration there is a separate value type.

Enumeration	Description
Activity	The presentity's activity (available, busy, lunch, etc.)
PlaceType	At what kind of place the presentity is (home, office, etc.)
Privacy	The amount of privacy the user wants (public, quiet, etc.)
Sphere	The user's current environment (work, home)
Communication	The user's means of communication (phone, mail, etc.)
Mood	The user"s mood (angry, confused, happy, etc.)
PlaceIs	Describes the properties of the place the user is currently at.
TimeOffset	Describes the number of minutes of offset from UTC that the user is currently at.
StatusIcon	Depicts the current status of the user.
Other	A name - value pair for arbitrary presence information

7.2 ActivityValue enumeration

This enumeration shows the user's current activity. If the activity is unknown, the attribute value will be `ActivityNone`, meaning the attribute was not set. If the user is doing something not in this list, the value will be set to `ActivityOther`.

Enumeration	Description
ActivityNone	Not set.
Appointment	The user has an appointment.
Available	The user is available for communication.
Busy	The user is busy and is only available for urgent matters.
DoNotDisturb	The user is very busy and does not wish to be disturbed.
OnThePhone	The user is on the phone.
Steering	The user is driving a car / train / airplane, etc.
Meeting	The user is in a meeting.
Away	No idea what the user is doing, but he is away.
Meal	The user is eating.
Breakfast	The user is having breakfast.
Lunch	The user is having lunch.
Dinner	The user is having dinner.
PermanentAbsence	The user is away and will not return for an extended period.
Vacation	The user is on vacation.
Holiday	A scheduled national or local holiday.
Performance	The user is in a theatre / concert.
InTransit	The user is in the transit area of an (air)port.
Travel	The user is travelling.
Sleeping	The user is sleeping.
LookingForWork	The user is looking for (paid) work.
Playing	The user is occupying him- or herself in amusement, sport, or other recreation.
Presentation	The user is giving a presentation, lecture, or participating in a formal round-table discussion.
Shopping	The user is visiting stores in search of goods or services.
Spectator	The user is observing an event, such as a sports event.
TV	The user is watching television.
Working	The user is engaged in, typically paid, labor, as part of a profession or job.
Worship	The user is participating in religious rites.
ActivityOther	The user is doing something not in this list.

7.3 PlaceTypeValue enumeration

This enumeration shows the type of the user's current location. If the place type is unknown, the attribute value will be `PlaceNone`, meaning the attribute was not set. If the user in a place not in this list, the value will be set to `PlaceOther`.

Enumeration	Description
PlaceNone	Not set.
Arena	The user is at an enclosed area used for sports events.
Home	The user is at home.
Office	The user is in an office.
PublicTransport	The user is on public transport.
Street	Walking on the street.
Outdoors	Generally outdoors.
PublicPlace	The user is in a public place.
Hotel	The user is in a hotel.
Theatre	The user is in a theatre or concert.
Restaurant	The user is in a restaurant, coffee shop or, other public dining establishment.
School	The user is at school.
Industrial	The user is in an industrial building.
Quiet	The user is in a quiet area.
Noisy	The user is in a noisy area.
Aircraft	The user is on an aircraft.
Watercraft	The user is on a vessel for travel on water such as a boat or ship.
Automobile	The user is in a car.
Bus	The user is in a bus.
BusStation	The user is in a bus- station.
TrainStation	The user is in a train-station.
ShoppingArea	The user is in a shopping mall or shopping area.
Airport	The user is in an airport.
Train	The user is in a train.
Bank	The user is in a bank.
Bar	The user is in a bar.
Bicycle	The user is on a bicycle.
Café	The user is in a café; usually a small and informal establishment that serves various refreshments (such as coffee); coffee shop.
Classroom	The user is in an academic classroom or lecture hall.
Club	The user is in a dance club, nightclub, or discotheque.
Construction	The user is at a contstruction site.
ConventionCenter	The user is in a convention center or exhibition hall.
Government	The user is in a government building, such as those used by the legislative, executive, or judicial branches of governments, including court houses, police stations, and military installations.
Hospital	The user is in a hospital, hospice, medical clinic, mental institution, or doctor's office.
Library	The user is in a library.
Motorcycle	The user is on a motorcycle.
Outdoors	The user outside a building, in or into the open air, such as a park or city streets.
Parking	The user is in a parking lot or parking garage.
PlaceOfWorship	The user is at a religious site where congregations gather for religious observances, such as a church, chapel, meetinghouse, mosque, shrine, synagogue, or temple.
Prison	The user is in a prison, penitentiary, jail or a brig.
Residence	The user is in a private or residential setting.
Stadium	The user is in a stadium.
Store	The user is in a shop or store.
Truck	The user is in a truck.
Underway	The user is in a land-, water-, or aircraft that is underway (in motion).
Warehouse	The user is in a warehouse.
Water	The user is in, on, or above bodies of water, such as an ocean, lake, river, canal, or other waterway.
PlaceOther	The user is in a kind of place not listed here.

7.4 PrivacyType enumeration

This enumeration shows the amount of privacy a user currently has. If the privacy is unknown, the attribute value will be `PrivacyNone`, meaning the attribute was not set. If none of the values apply, the value will be set to `PrivacyOther`.

Enumeration	Description
PrivacyNone	Not set.
PrivacyPublic	The user is surrounded by other people and cannot discuss openly.
PrivacyPrivate	The user is alone and able to talk openly.
PrivacyQuiet	The user is in a quiet environment and cannot talk at all.
PrivacyOther	None of the other values applies.
PrivacyAudio	Inappropriate individuals are not likely to overhear audio communications.
PrivacyText	Inappropriate individuals are not likely to see text communications.
PrivacyVideo	Inappropriate individuals are not likely to see video communications.

7.5 SphereValue enumeration

This enumeration shows the sphere within which the user acts. If the sphere is unknown, the attribute value will be `SphereNone`, meaning the attribute was not set. If the sphere is not in this list (neither work nor home), the value will be set to `SphereOther`.

Enumeration	Description
SphereNone	Not set.
SphereWork	The user is acting within his work sphere, i.e. as a member of his company
SphereHome	The user is acting within his home sphere, i.e. as a private person.
SphereOther	The user is acting neither within his work nor within his home sphere.

7.6 CommunicationMeansType enumeration

This enumeration lists communication means. If the communication attribute referrers to a means not in this list, it will point to `MeansOther`.

Enumeration	Description
Phone	The communication attribute refers to a phone (fixed line or mobile or SIP).
Chat	The communication attribute refers to a chat client.
SMS	The communication attribute refers to an SMS client.
Video	The communication attribute refers to a video phone (fixed line or mobile or SIP).
Web	The communication attribute refers to a web client.
EMail	The communication attribute refers to an e-mail client.
MMS	The communication attribute refers to an MMS client.
MeansOther	The communication attribute refers to any other client.

7.7 CommunicationMeans structure

This structure describes one way of reaching the presentity or contacts of the presentity. If the presentity for example is unavailable he/she may publish communication means for one of his/her contacts.

Element name	Element type	Optional	Description
Priority	xsd:float	No	The priority of this communication means. Between 0 and 1, the latter meaning the highest priority.
Contact	xsd:anyURI	No	The contact address for this communication means.
Type	CommunicationMeansType	No	The type of this communication means.
Status	CommunicationStatusType	Yes	The status of this communication means.
Relationship	RelationshipValue	Yes	Relationship the presentity has with the contact whose communication means address is published. If the parameter is not set it means that the communication means refers to the presentity himself.

7.8 CommunicationValue structure

This structure describes the various ways of reaching a presentity.

Element name	Element type	Optional	Description
Means	CommunicationMeans [0..unbounded]	Yes	The different ways of reaching the presentity.

7.9 OtherValue structure

This structure can be used for storing arbitrary data about a presentity.

Element name	Element type	Optional	Description
Name	xsd:string	No	Description of the content.
Value	xsd:string	No	Attribute content.

7.10 PresenceAttribute structure

Presence data published by a presentity and retrieved by watchers.

Element name	Element type	Optional	Description
LastChange	xsd:dateTime	No	The time and date when the attribute was changed last.
Note	xsd:string	Yes	An explanatory note.
TypeAndValue	AttributeTypeAndValue	No	The type of attribute and its associated value.

7.10a AttributeTypeAndValue union

Element name	Element type	Optional	Description
UnionElement	PresenceAttributeType	No	Type of presence attribute provided (one of the following).
Activity	ActivityValue	Yes	The presentity's activity (available, busy, lunch, etc.)
PlaceType	PlaceTypeValue	Yes	At what kind of place the presentity is (home, office, etc.)
Privacy	PrivacyValue	Yes	The amount of privacy the user wants (public, quiet, etc.)
Sphere	SphereValue	Yes	The user's current environment (work, home)
Communication	CommunicationValue	Yes	The user's means of communication (phone, mail, etc.)
Mood	MoodValue	Yes	The user"s mood (angry, confused, happy, etc.)
PlaceIs	PlaceIsValue	Yes	Describes the properties of the place the user is currently at.
TimeOffset	TimeOffsetValue	Yes	Describes the number of minutes of offset from UTC that the user is currently at.
StatusIcon	StatusIconValue	Yes	Depicts the current status of the user.
Other	OtherValue	Yes	A name - value pair for arbitrary presence information

7.11 SubscriptionRequest structure

This structure is returned to the presentity by the Presence Web Service and contains the requesting watcher and the attributes he wants to subscribe.

Element name	Element type	Optional	Description
Watcher	xsd:anyURI	No	The watcher who wants to gain access to data.
Attributes	PresenceAttributeType [0..unbounded]	Yes	The attributes the watcher wants to see. An empty array means subscription to all attribute types.
Application	xsd:string	No	The name of the application running on behalf of the watcher. Note that this field has solely informative purposes, access rights management is based on watcher id only.

7.12 Void

7.13 CommunicationStatusType enumeration

This enumeration shows the status of communication means.

Enumeration	Description
On	Presentity has his own communication means that is available now. Watcher can connect directly.
Off	Presentity has his own communication means that is not available for some reason.
Busy	Presentity has his own communication means that is busy.

7.14 PrivacyValue structure

This structure holds an array of privacy types. A privacy type may indicate whether other parties are likely to observe a specific communication type.

Element name	Element type	Optional	Description
PrivacyTypes	PrivacyType [1… unbounded]	No	Array holding privacy types.

7.15 MoodValue enumeration

This enumeration shows the type of the user's current mood. If the mood type is unknown, the attribute value will be `MoodNone`, meaning the attribute was not set. If the mood type is not in this list, the value will be set to `MoodOther`.

Enumeration	Description
Afraid	The user is afraid.
Amazed	The user is amazed.

Angry	The user is angry.
Annoyed	The user is annoyed.
Anxious	The user is anxious.
Ashamed	The user is ashamed.
Bored	The user is bored.
Brave	The user is brave.
Calm	The user is calm.
Cold	The user is cold.
Confused	The user is confused.
Contented	The user is contented.
Cranky	The user is cranky.
Curious	The user is curious.
Depressed	The user is depressed.
Disappointed	The user is disappointed.
Disgusted	The user is disgusted.
Distracted	The user is distracted.
Embarrassed	The user is embarrassed.
Excited	The user is excited.
Flirtatious	The user is flirtatious.
Frustrated	The user is frustrated.
Grumpy	The user is grumpy.
Guilty	The user is guilty.
Happy	The user is happy.
Hot	The user is hot.
Humbled	The user is humbled.
Humiliated	The user is humiliated.
Hungry	The user is hungry.
Hurt	The user is hurt.
Impressed	The user is impressed.
InAwe	The user is in awe.
InLove	The user is in love.
Indignant	The user is indignant.
Interested	The user is interested.
Invincible	The user is invincible.
Jealous	The user is jealous.
Lonely	The user is lonely.
Mean	The user is mean.
MoodNone	The user"s mood is unknown.
Moody	The user is moody.
Nervous	The user is nervous.
Neutral	The user is neutral.
Offended	The user is offended.
Playful	The user is playful.
Proud	The user is proud.
Relieved	The user is relieved.
Remorseful	The user is remorseful.
Restless	The user is restless.
Sad	The user is sad.
Sarcastic	The user is sarcastic.
Serious	The user is serious.
Shocked	The user is shocked.
Shy	The user is shy.
Sick	The user is sick.
Sleepy	The user is sleepy.
Stressed	The user is stressed.
Surprised	The user is surprised.
Thirsty	The user is thirsty.
Worried	The user is worried.
MoodOther	The user"s current mood is not listed here.

7.16 PlaceIsValue structure

This structure holds properties of the place the presentity is currently at, such as the levels of light and noise. This information can be used by the watcher to determine the type of communication that is likely to be successful.

Element name	Element type	Optional	Description
Audio	PlaceIsAudioValue	Yes	Describes place conditions for audio communication.
Video	PlaceIsVideoValue	Yes	Describes place conditions for video communication.
Text	PlaceIsTextValue	Yes	Describes place conditions for real-time and instant-messaging communication.

7.17 PlaceIsAudioValue enumeration

This enumeration shows the properties of the place the presentity is currently at with respect to audio communication.

Enumeration	Description
Noisy	The user is in a place with a level of background noise that makes audio communications difficult.
Ok	The environmental conditions are suitable.
Quiet	The user is in a place such as a library, restaurant, place of worship, or theater that discourages noise, conversation, and other distractions.
Unknown	The place attributes are not known.

7.18 PlaceIsVideoValue enumeration

This enumeration shows the properties of the place the presentity is currently at with respect to video communication.

Enumeration	Description
TooBright	The place is too bright for video communication.
Ok	The environmental conditions for video communication are acceptable.
Dark	The place is too dark for video communication.
Unknown	The environmental conditions for video communication are not known.

7.19 PlaceIsTextValue enumeration

This enumeration shows the properties of the place the presentity is currently at with respect to real-time text and instant messaging.

Enumeration	Description
Uncomfortable	The place is uncomfortable for typing or other text entry.
Inappropriate	The place is inappropriate for typing or other text entry.
Ok	The environmental conditions are suitable for typing or other text entry.
Unknown	The place attributes for text communication is not known.

7.20 RelationshipValue enumeration

This enumeration shows the type of relationship the user has with a contact. If the relationship type is unknown, the attribute value will be `RelationshipNone`, meaning the attribute was not set. If the relationship type is not in this list, the value will be set to `RelationshipOther`.

Enumeration	Description
Family	The contact is part of the user"s family.
Assistant	The contact is an assistant of the user, e.g. colleague.
Friend	The contact is a friend of the user.
Associate	The contact is an associate of the user.
Supervisor	The contact is the user"s supervisor.
RelationshipNone	The relationship type is unknown.
RelationshipOther	The relationship type is not in this list.

7.21 TimeOffsetValue structure

This structure describes the number of minutes of offset from UTC that the user is currently at.

Element name	Element type	Optional	Description
TimeZone	xsd:string	Yes	Describes the time zone. The description is meant for human presentation.
Minutes	xsd:int	No	Number of minutes of offset from UTC that the user is currently at.

7.22 StatusIconValue structure

This structure includes a URI pointing to an image that represents the current status of the user.

Element name	Element type	Optional	Description
StatusIcon	xsd:anyURI	No	Address to the image (icon) representing the user"s current status.

7.23 Watcher structure

This structure holds a watcher identity, the subscription status and trigger event which caused the transition to the current status.

Element name	Element type	Optional	Description
Watcher	xsd:anyURI	No	The identity of the watcher.
WatcherSubscriptionStatus	WatcherSubscriptionStatus	No	The status of the watcher"s subscription.
SubscriptionStatusTriggerEvent	SubscriptionStatusTriggerEvent	No	The event that caused the transition to this status.

7.24 WatcherSubscriptionStatus enumeration

This enumeration shows the different statuses a watcher"s subscription may have.

Enumeration	Description
Authorised	Indicates that the watcher"s subscription has been authorised by the presentity
Blocked	Indicates that the watcher"s subscription is currently blocked by the presentity.
PoliteBlocked	Indicates that the watcher"s subscription is politely blocked by the presentity.
Pending	Indicates that the watcher"s subscription is awaiting an authorization decision
Active	Indicates that the watcher"s subscription is active and has been approved by the presentity.
Waiting	Indicates that the watcher"s subscription has timed out whilst still awaiting an authorization decision.
Terminated	Indicates that the watcher"s subscription has been terminated. The Subscription Status Trigger Event gives an indication of the event that caused the termination of the subscription.

7.25 SubscriptionStatusTriggerEvent enumeration

This enumeration shows the different events that cause a transition in the status of a watcher"s subscription.

Enumeration	Description
Subscribe	The subscription has arrived and the policy existing at the time of arrival has been applied.
Approved	The subscription has been approved by the presentity.
Deactivated	The server has discarded a subscription (with no change in authorization policy).
Probation	The server has terminated a subscription and requested that the subscriber waits before retrying.
Rejected	The subscription has been rejected by the presentity.
Timeout	The subscription has timed out.
Giveup	A pending or waiting subscription has been terminated.

7.26 Presence Notification structure

This strucrure holds the presentity, subscription status and presence data that are notified. The presence data shall be notified in the form of the person, service and device attributes.

Element name	Element type	Optional	Description
Presentity	xsd:anyURI	No	The presentity whose presence status is notified.
WatcherSubscriptionStatus	WatcherSubscriptionStatus	Yes	The status of the subscription to the presentity.
NotifiedPersonAttributes	PresenceAttribute [0..unbounded]	Yes	The presence data related to a person.
NotifiedServiceAttributes	PresenceAttribute [0..unbounded]	Yes	The presence data related to particular services.
NotifiedDeviceAttributes	PresenceAttribute [0..unbounded]	Yes	The presence data related to particular devices.

7.27 Validity structure

This structure describes the time boundaries during which the information subject to this element is valid.

Element name	Element type	Optional	Description
From	xsd:dateTime	No	The time and date when the time boundary begins.
Until	xsd:dateTime	No	The time and date when the time boundary ends.

7.28 Authorization Value enumeration

This enumeration shows the different authorization actions the presentity may apply to a watcher.

Enumeration	Description
Block	Indicates that the watcher is blocked by the presentity.
Confirm	Indicates that the watcher shall await input from the presentity before it is determined how the watcher shall be allowed to proceed.
PoliteBlock	Indicates that the watcher is politely blocked by the presentity.
Allow	Indicates that the watcher is allowed by the presentity.

7.29 DevicesProvided structure

This structure is used to indicate what devices the presentity wants provided to the watcher. The result shall be a union of all the elements in the structure.

Element name	Element type	Optional	Description
All	xsd:boolean	Yes	If true, all devices that are present are provided.
Class	xsd:string	Yes	Class of device if present, to be provided.
OccurrenceId	xsd:string	Yes	Occurrence Id of device if present, to be provided.
DeviceId	xsd:string	Yes	Device Id of device if present, to be provided.

7.30 PersonsProvided structure

This structure is used to indicate what persons (sets of person information) the presentity wants provided to the watcher. The result shall be a union of all the elements in the structure.

Element name	Element type	Optional	Description
All	xsd:boolean	Yes	If true, all persons that are present are provided.
Class	xsd:string	Yes	Class of person if present, to be provided.
OccurrenceId	xsd:string	Yes	Occurrence Id of person if present, to be provided.

7.31 ServicesProvided structure

This structure is used to indicate what services the presentity wants provided to the watcher. The result shall be a union of all the elements in the structure.

Element name	Element type	Optional	Description
All	xsd:boolean	Yes	If true, all services that are present are provided.
Class	xsd:string	Yes	Class of service if present, to be provided.
OccurrenceId	xsd:string	Yes	Occurrence Id of service if present, to be provided.
ServiceUri	xsd:anyURI	Yes	Service-uri of service if present, to be provided.
ServiceUriScheme	xsd:string	Yes	Service-uri-scheme of service if present, to be provided.
ServiceId	xsd:string	Yes	Service Identity of service if present, to be provided.

7.32 AttributePermission structure

The presentity"s permissions for particular presence attributes to be provided to the watcher for allowed device(s), person(s) or service(s).

Element name	Element type	Optional	Description
Attribute	PresenceAttributeType	No	The name of the attribute type.
Decision	xsd:Boolean	No	Indicates whether the presentity provides the attribute type (if present), (true) or not (false).

7.33 PresenceDataFormat enumeration

This enumeration shows the presence data format that may be requested.

Only one value – PIDF - is currently defined.

Enumeration	Description
PIDF	PIDF document – see RFC 3863 [16] and any extensions (for example, the extensions to PIDF defined in RFC 4480 [17]).

8 Web Service interface definition

This API is separated into three interfaces:

- PresenceConsumer interface: watcher methods for requesting and subscribing presence data.
- PresenceNotification interface: is the watcher notification interface for presence events.
- PresenceSupplier interface: presentity methods for supplying presence data and managing subscriptions.

8.1 Interface: PresenceConsumer

Client role: watcher.

This set of methods is used by the watcher to obtain presence data. The watcher can select between a polling mode or a notification mode in order to receive presence data.

8.1.1 Void

8.1.2 Operation: getUserPresenceRequest

Returns the aggregated presence data of a presentity or a group / list of Presentities to the watcher. Only the attributes which the watcher is entitled to see will be returned..

The Attributes message part specifies a subset of all possible attribute types that can be returned and can be used as a filter. If the Attributes message part is empty, the watcher wants to be notified of all attribute types.

If the request identifies a group / list of multiple Presentities, the watcher may receive presence attributes for every Presentity in the group / list.

Note: A mapping to SIP / IMS networks for multiple Presentities may involve the routing of the request to an RLS Server rather than directly to a Presence Server.

The returned attributes will be dependent on the authorization afforded to the watcher by each presentity.

The "Waiting" value of Watcher Subscription Status allows a Presentity to be informed of a request that expired without an authorization decision. "Waiting" subscriptions can be reported when the presentity subscribes to watcher information via getMyWatchers() or / and getOpenSubscriptions(). The presentity may then authorize the watcher (via updateSubscriptionAuthorization() before the request from the watcher is repeated.

8.1.2.1 Input message: getUserPresenceRequest

Part name	Part type	Optional	Description
watcher	xsd:anyURI	No	The watcher who wants to see a presentity or a group / list of presentities. The Watcher Application invokes this operation on behalf of this watcher. It is assumed that the Watcher Application has authenticated the watcher.
Presentities	xsd:anyURI [1 … unbounded]	No	The presentity or group / list of presentities whose attributes the watcher wants to see. Note: In the latter case, the network may only allow the list owner to subscribe to a list of presentities. i.e. the watcher may have to be the list owner.
MultiplePresentities	xsd:Boolean	Yes	The watcher accepts notifications for multiple Presentities.
Attributes	PresenceAttributeType [0..unbounded]	Yes	The attribute types the watcher wants to see (The same attributes for all list members). An empty array means all attribute types.
Application	xsd:string	Yes	Describes the application the watcher needs the data for. (Operator Policy may make this parameter mandatory).
Format	PresenceDataFormat	Yes	If present, the watcher accepts Presence data formatted as described e.g. as a PIDF document – see RFC 3863 [16] and any extensions (for example, the extensions to PIDF defined in RFC 4480 [17]).

8.1.2.2 Output message: getUserPresenceResponse

In the Output message, one of "result", "NotificationResult" or "PresenceDocument" must be provided. Only one should be provided.

Part name	Part type	Optional	Description
result	PresenceAttribute [0..unbounded]	Yes	The actual presence data.
NotificationResult	PresenceNotification [0..unbounded]	Yes	The presence data in a Presence Notification structure form.
PresenceDocument	xsd:string	Yes	Presence data formatted as a PIDF document in RFC 3863 [16] and any extensions (for example, the extensions to PIDF defined in RFC 4480 [17]).

8.1.2.3 Referenced faults

ServiceException from 3GPP TS 29.199-1 [6]:

- SVC0001: Service error.
- SVC0002: Invalid input value.
- SVC0004: No valid addresses - if the presentity address does not exist.

PolicyException from 3GPP TS 29.199-1 [6]. The presentity has the possibility to cancel or block a subscription by manipulating the policy rules. The exception informs the watcher about this status change:

- POL0001: Policy error.
- POL0002: Privacy error.

8.1.3 Operation: startPresenceNotification

Initiates a subscription for notifications of the presence of a Presentity, or of the presence of a group / list of Presentities, to the watcher via statusNotified().

NOTE: A mapping to SIP / IMS networks for multiple Presentities may involve the routing of the request to an RLS Server rather than directly to a Presence Server.

The Reference message part contains the Web Service reference that provides the information necessary for the Presence Supplier to notify the watcher with the results of the subscription request. The correlator element of this Web service reference associates the notification events in subsequent messages with the request.

The Attributes message part specifies a subset of all possible attribute types that can be subscribed and can be used as a filter. The watcher sets a notification trigger on certain user presence attribute changes. If the Attributes message part is empty, the watcher wants to be notified about changes to all attribute types.

If the request identifies a list of multiple Presentities, the watcher will receive notifications for every Presentity in the list. However, the notified attributes will be dependent on the watcher subscription status for each presentity.

For each presentity, limited or no presence information may be present in the resulting notifications until the presentity has authorized the watcher and allowed access to attributes.

In order to be informed of the subscription and authorize a new watcher (via updateSubscriptionAuthorization(), the presentity can subscribe to watcher information via getMyWatchers() or / and getOpenSubscriptions().

8.1.3.1 Input message: startPresenceNotificationRequest

Part name	Part type	Optional	Description
watcher	xsd:anyURI	No	The watcher who wants to monitor a presentity or a group / list of presentities. The Watcher Application invokes this operation on behalf of this watcher. It is assumed that the Watcher Application has authenticated the watcher.
Presentities	xsd:anyURI [1 … unbounded]	No	The presentity or group / list of presentities whose attributes the watcher wants to monitor . Note: In the latter case, a network may only allow the list owner to subscribe to a list of Presentities i.e. the watcher may have to be the list owner.
MultiplePresentities	xsd:Boolean	Yes	If present and true, the watcher accepts notifications for multiple Presentities.
Attributes	PresenceAttributeType [0..unbounded]	Yes	The attribute types the watcher wants to access. (The same attributes for all list members). An empty array means monitoring of all attribute types.
Application	xsd:string	Yes	Describes the application the watcher needs the data for. (Operator Policy may make this parameter mandatory).
Reference	common:SimpleReference	No	The notification interface (contains the correlator string used in subsequent messages to the notification interface).
Frequency	common:TimeMetric	Yes	Maximum frequency of notifications (can also be considered minimum time between notifications). In case of a group subscription the service must make sure this frequency is not violated by notifications for various members of the group, especially in combination with `checkImmediate`.
Duration	common:TimeMetric	Yes	Length of time notifications occur for, do not specify to use default notification time defined by service policy.
Count	xsd:int	Yes	Maximum number of notifications. For no maximum, either do not specify this part or specify a value of zero.
CheckImmediate	xsd:boolean	Yes	Whether to check status immediately after establishing notification. If absent, the default action is as if set to "true".
Format	PresenceDataFormat	Yes	If present, the watcther accepts Presence data formatted as described e.g. as a PIDF document – see RFC 3863 [16] and any extensions (for example, the extensions to PIDF defined in RFC 4480 [17]).

8.1.3.2 Output message: startPresenceNotificationResponse

Part name	Part type	Optional	Description
result	xsd:anyURI [0..unbounded]	Yes	The presentities for which the requested notifications could not be set up. Empty if notifications were set up for all the specified Presentities or the network does not return this information.

8.1.3.3 Referenced faults

ServiceException from 3GPP TS 29.199-1 [6]:

- SVC0001: Service error.
- SVC0002: Invalid input value.
- SVC0004: No valid addresses - if the presentity address does not exist.
- SVC0005: Duplicate correlator.
- SVC0006: Invalid Group

PolicyException from 3GPP TS 29.199-1 [6]. The presentity has the possibility to cancel or block a subscription by manipulating the policy rules. The exception informs the watcher about this status change:

- POL0001: Policy error.
- POL0002: Privacy error.
- POL0003: Too many addresses.
- POL0004: Unlimited notifications not supported
- POL0005: Too many notifications requested.
- POL0006: Groups not allowed.
- POL0007: Nested groups not allowed.
- POL0009: Invalid frequency requested.

8.1.4 Operation: endPresenceNotificationRequest

Indicates that the watcher does not want further notifications for a specific notification request (identified by the correlator).

8.1.4.1 Input message: endPresenceNotificationRequest

Part name	Part type	Optional	Description
Correlator	xsd:string	No	The notification the watcher wants to cancel.

8.1.4.2 Output message: endPresenceNotificationResponse

Part name	Part type	Optional	Description
None			

8.1.4.3 Referenced faults

ServiceException from 3GPP TS 29.199-1 [6]:

- SVC0001: Service error.
- SVC0002: Invalid input value.

PolicyException from 3GPP TS 29.199-1 [6]:

- POL0001: Policy error.

8.2 Interface: PresenceNotification

This client callback interface is used by the presence consumer interface to send notifications.

8.2.1 Void

8.2.2 Operation: statusEnd

The notifications have ended for this correlator. This message will be delivered when the duration or count for notifications have been completed. This message will not be delivered in the case of an error ending the notifications or deliberate ending of the notifications (using endPresenceNotification operation).

8.2.2.1 Input message: statusEndRequest

Part name	Part type	Optional	Description
Correlator	xsd:string	No	Correlator provided in request to set up this notification

8.2.2.2 Output message: statusEndResponse

Part name	Part type	Optional	Description
None			

8.2.2.3 Referenced faults

None.

8.2.3 Void

8.2.4 Operation: subscriptionEnded

This asynchronous operation is called by the Web Service to notify the watcher (application) that the subscription has terminated. Typical reasons are a timeout of the underlying SIP soft state subscription (in accordance with [14] and [9]) or the decision of the presentity to block further presence information to that watcher.

8.2.4.1 Input message: subscriptionEndedRequest

Part name	Part type	Optional	Description
correlator	xsd:string	No	Identifies the notification request.
Presentity	xsd:anyURI	No	The presentity (or list owner) to which the subscription has terminated
Watcher	xsd:anyURI	Yes	The watcher whose subscription is terminated. It has the same value as the watcher part of the original subscribePresenceRequest message (reference clause 8.1.1.1).
Reason	xsd:string	No	Timeout, Blocked, No Resource, Give Up, Probation, Deactivated.

8.2.4.2 Output message: subscriptionEndedResponse

Part name	Part type	Optional	Description
None			

8.2.5 Operation: statusNotifiedRequest

This asynchronous operation is called by the Web Service as a result of the handling of a startPresenceNotification() request from a watcher application or / and when an attribute for which notifications were requested changes.

8.2.5.1 Input message: statusNotifiedRequest

In the Output message, one of "PresenceInformation" or "PresenceDocument" must be provided. Only one should be provided.

Part name	Part type	Optional	Description
correlator	xsd:string	No	Identifies the notification request.
PresenceInformation	PresenceNotification [0..unbounded]	Yes	The presence data notified for each presentity.
PresenceDocument	xsd:string	Yes	Presence data formatted as a PIDF document as in RFC 3863 [16] and any extensions (for example, the extensions to PIDF defined in RFC 4480 [17]).

8.2.5.1 Output message: statusNotifiedResponse

Part name	Part type	Optional	Description
None			

8.2.5.3 Referenced faults

None

8.3 Interface: PresenceSupplier

These methods are used by the presentity to supply presence data and manage access to the data by its watchers. We assume that the presentity has been previously authenticated, so that his Identity is known.

8.3.1 Operation: publish

The presentity publishes data about herself. This data will then be filtered by the system and forwarded to the watchers who have ordered notifications. Either the Presence Document as a PIDF document shall be published or the presence data shall be published in the form of either generic presence attributes or via person, service and device attributes.

8.3.1.1 Input message: publishRequest

Part name	Part type	Optional	Description
presentity	xsd:anyURI	No	The presentity who wants to publish his or her presence data. The Presentity Application invokes this operation on behalf of this presentity. It is assumed that the Presentity Application has authenticated the presentity.
Presence	PresenceAttribute [0..unbounded]	Yes	The presence attributes the devices of the presentity supports
PublishedPersonAttributes	PresenceAttribute [0..unbounded]	Yes	The presence data related to a person.
PublishedServiceAttributes	PresenceAttribute [0..unbounded]	Yes	The presence data related to particular services.
PublishedDeviceAttributes	PresenceAttribute [0..unbounded]	Yes	The presence data related to particular devices.
PresenceDocument	PIDFDocument	Yes	Presence data formatted as a PIDF document – see RFC 3863 [16] and any extensions (for example, the extensions to PIDF defined in RFC 4480 [17]).

8.3.1.2 Output message: publishResponse

Part name	Part type	Optional	Description
None			

8.3.1.3 Referenced faults

ServiceException from 3GPP TS 29.199-1 [6]:

- SVC0001: Service error.
- SVC0002: Invalid input value.

PolicyException from 3GPP TS 29.199-1 [6]:

- POL0001: Policy error.

8.3.2 Void

8.3.3 Void

8.3.4 Operation: getMyWatchersRequest

This operation may be called periodically by the presentity (application) to see any watcher(s) subscribing or wanting to subscribe to presence data.

Returns an array of watching identities for a specific presentity or a structure that includes the identity of watching subscribers for the presentity. The watchers" subscription status and the event that caused the transition to this status are included in the structure. The client may apply a filter to retrieve watchers based on subscription status.

The client can answer open requests (i.e. those with a Watcher Subscription Status of "Pending" or "Waiting") with updateSubscriptionAuthorization().

8.3.4.1 Input message: getMyWatchersRequest

Part name	Part type	Optional	Description
presentity	xsd:anyURI	No	Presentity who wants to know his or her watchers. The Presentity Application invokes this operation on behalf of this presentity. It is assumed that the Presentity Application has authenticated the presentity.
WatcherSubscriptionStatus	WatcherSubscriptionStatus [0 … unbounded]	Yes	Allows the presentity to apply a filter based on watcher subscription status. An empty list means the presentity wants to retrieve all watchers.

8.3.4.2 Output message: getMyWatchersResponse

Part name	Part type	Optional	Description
result	Watcher [0..unbounded]	Yes	The list of watcher identities that currently have requested access to the presentity's attributes, The subscription status of each watcher is included in the response and the event that caused the transition to the current status.

8.3.4.3 Referenced faults

ServiceException from 3GPP TS 29.199-1 [6]:

- SVC0001: Service error.
- SVC0002: Invalid input value.
- SVC0004: No valid addresses - if the presentity address does not exist.

PolicyException from 3GPP TS 29.199-1 [6]:

- POL0001: Policy error.
- POL0002: Privacy error.

8.3.5 Operation: getSubscribedAttributes

Returns an array of attributes that a specific watcher has subscribed.

8.3.5.1 Input message: getSubscribedAttributesRequest

Part name	Part type	Optional	Description
presentity	xsd:anyURI	No	The presentity who wants to know the attributes to which his or her watcher has subscribed. The Presentity Application invokes this operation on behalf of this presentity. However, it should NOT be assumed that the Presentity Application has authenticated the presentity.
Watcher	xsd:anyURI	No	The watcher whose subscriptions the presentity wants to know

8.3.5.2 Output message: getSubscribedAttributesResponse

Part name	Part type	Optional	Description
Result	PresenceAttributeType [0..unbounded]	Yes	The attributes the watcher is subscribed to. An empty array means subscription to all attribute types.

8.3.5.3 Referenced faults

ServiceException from 3GPP TS 29.199-1 [6]:

- SVC0001: Service error.
- SVC0004: No valid addresses.
- SVC0221: Not a watcher - if the URI in the field watcher is not a watcher of the presentity.

PolicyException from 3GPP TS 29.199-1 [6]:

- POL0001: Policy error.

8.3.6 Void

8.3.7 Operation: updateAuthorizationRuleRequest

The presentity may use this operation in advance of watcher requests or in response to watcher requests for which no authorization policy exists.

The operation can be used by the presentity at anytime to change the authorization for certain watchers.

The operation contains a rule consisting of watcher or prospective watcher identities along with the authorization decisions and attribute permissions applicable to them.

If the optional "Rules" (XML Element) part is provided, none of the other optional elements are provided and the only other parameter necessary is the "presentity". The "Rules" XML Element may contain single or multiple rules (including rule ids) which are additive to those provided in previous requests.

If there is a conflict in the rules, the outcome is decided by the priority rules for authorization decisions in the network (e.g. "Allow" may take first priority followed by "Polite Block", then "Confirm" then .Block").

If the optional "Rules" (XML Element) part is NOT provided, the optional "RuleId" and "AuthorizationDecision" shall always be provided.

Watchers may be identified singly or by membership of a domain or list. When membership of a domain or list is used to identify watchers, exceptions within the domain or list for which the rule does not apply may be identified via the "Except Watchers", "Except Domain" or "Except List" parameters.

It is also possible for the rule to be applied to Anonymous, Other (not specified by other rules) or Any Watchers.

The rule only applies when the current time is within the range specified by the Validity parameter.

The permissions are "additive" i.e. the overall permissions granted via multiple rules matching the watcher are the union of the permissions in all the rules. Rules provide (albeit limited) permissions and do not deny them. Thus removing a rule does not increase permissions.

The permissions are hierarchical i.e. devices, persons and services are only provided to authorized subscriptions and attribute rules only apply to the devices, persons and services that are provided.

When the authorization decision is "Polite Block", the watcher is not considered authorised by the presentity, but this information is not disclosed to the watcher. Instead, the watcher"s request is accepted, but the watcher receives no presence information or only that presence information dedicated for the polite block feature.

8.3.7.1 Input message: updateAuthorizationRuleRequest

Part name	Part type	Optional	Description
presentity	xsd:anyURI	No	Presentity who wants to update his or her authorization rules. The Presentity Application invokes this operation on behalf of this presentity. It is assumed that the Presentity Application has authenticated the presentity.
RuleId	xsd:string	No	An identifier for the rule that this request establishes or modifies. Note: Rules provide (albeit limited) permissions and do not deny them.
Watchers	xsd:anyURI [0... unbounded]	Yes	The watchers for whom the presentity wants to apply this rule..
Domains	xsd:string [0... unbounded]	Yes	List of domain names that contain watchers for whom the presentity wants to apply this rule.
ListNames	xsd:string [0... unbounded]	Yes	The names of URI resource lists owned by the presentity that contain watchers for whom the presentity wants to apply this rule.
AnonymousWatcher	xsd:boolean	Yes	If AnonymousWatcher is true, the request applies to anonymous watchers.
OtherWatcher	xsd:boolean	Yes	If OtherWatcher is true, the request applies to all watchers that are not identified in other valid policies.
AnyWatcher	xsd:boolean	Yes	If AnyWatcher is true, the request applies to any watcher.
ExceptWatchers	xsd:anyURI [0... unbounded]	Yes	The watchers whom the presentity does not want this rule to apply to.
ExceptDomains	xsd:string [0... unbounded]	Yes	List of domain names that contain watchers for whom the presentity does not want to apply this rule.
ExceptListNames	xsd:string [0... unbounded]	Yes	The names of URI resource lists owned by the presentity that contain watchers for whom the presentity does not want to apply this rule.
Validity	Validity	Yes	This policy is only applied whilst the current time is within the time range specified by this element. If this element is absent, the policy is permanently applied.
AuthorizationDecision	AuthorizationValue	Yes	Indicates what action the presentity wants applied to matching subscriptions.
DevicesProvided	DevicesProvided	Yes	Indicates which of his or her devices, the presentity wants provided to the watcher. If absent, no devices are provided.
PersonsProvided	PersonsProvided	Yes	Indicates which of his or her persons (sets of person information), the presentity wants provided to the watcher. If absent, no person information is provided.
ServicesProvided	ServicesProvided	Yes	Indicates which of his or her services, the presentity wants provided to the watcher. If absent, no services are provided.
AttributeRules	AttributePermission [0..unbounded]	Yes	The set of permissions defining which particular presence attributes are provided to allowed watchers that have device(s, person(s) or services(s) provided. An empty array means only the set of information that is always reported for allowed device(s, person(s) or services(s) is provided. The contact, service-class, basic status and timestamp elements are provided for all services, if present. The timestamp element is provided for all persons, if present. The timestamp and deviceID elements are provided for all devices, if present.
Rules	XML Element	Yes	XML element conforming to the OMA schema namespace: urn:oma:xml:prs:pres-rules (which extends the ietf nameplace: urn:ietf:params:xml:ns:pres-rules) [XSD-PRESRULES] [18].

8.3.7.2 Output message: updateAuthorizationRuleResponse

Part name	Part type	Optional	Description
None			

8.3.7.3 Referenced faults

ServiceException from 3GPP TS 29.199-1 [6]:

- SVC0001: Service error.
- SVC0002: Invalid input value.
- SVC0004: No valid addresses - if the presentity address does not exist.

PolicyException from 3GPP TS 29.199-1 [6]:

- POL0001: Policy error.
- POL0002: Privacy error.
- POL0003: Too many addresses.

8.3.8 Operation: deleteAuthorizationRuleRequest

Indicates that the presentity wishes to delete the authorization rule identified by the rule identifier.

8.3.8.1 Input message: deleteAuthorizationRuleRequest

Part name	Part type	Optional	Description
presentity	xsd:anyURI	No	Presentity who wants to delete the authorization rule. The Presentity Application invokes this operation on behalf of this presentity. It is assumed that the Presentity Application has authenticated the presentity.
RuleId	xsd:string	No	The identifier for the rule that this request deletes.

8.3.8.2 Output message: deleteAuthorizationRuleResponse

Part name	Part type	Optional	Description
None			

8.3.8.3 Referenced faults

ServiceException from 3GPP TS 29.199-1 [6]:

- SVC0001: Service error.
- SVC0002: Invalid input value.
- SVC0004: No valid addresses - if the presentity address does not exist.

PolicyException from 3GPP TS 29.199-1 [6]:

- POL0001: Policy error.

8.4 Interface: PresenceSupplierNotificationManager

This interface enables applications to set up and tear down notifications for watcher subscriptions.

8.4.1 Void

8.4.2 Void

8.4.3 Operation: startMyWatchersNotificationRequest

The PresenceSupplierNotificationManager provides functionality that allows presentities to be notified about all watcher subscriptions.

The client may apply a filter to retrieve watchers based on subscription status.

The application may specify the duration for how long the notifications should occur with the Duration parameter. If the parameter is absent the notification will last until the application invokes endMyWatchersNotification.

The correlator provided in the reference must be unique for the application at the time the notification is initiated, otherwise a ServiceException (SVC0005) will be returned to the application.

8.4.3.1 Input message: startMyWatchersNotificationRequest

Part name	Part type	Optional	Description
reference	common:SimpleReference	No	Notification endpoint definition.
Presentity	xsd:anyURI	No	The presentity for which the application wants to know the watchers.
WatcherSubscriptionStatus	WatcherSubscriptionStatus [0 … unbounded]	Yes	Allows a filter to be applied based on watcher subscription status. An empty list means retrieve all watchers.
Duration	common:TimeMetric	Yes	Length of time notifications occur for. If the element is omitted, the notifications will continue until endMyWatchersNotification is invoked.

8.4.3.2 Output message: startMyWatchersNotificationResponse

Part name	Part type	Optional	Description
None			

8.4.3.3 Referenced faults

ServiceException from 3GPP TS 29.199-1 [6]:

- SVC0001: Service error.
- SVC0002: Invalid input value.
- SVC0004: No valid addresses - if the presentity address does not exist.
- SVC0005: Duplicate correlator.
- SVC0008: Overlapping criteria.

PolicyException from 3GPP TS 29.199-1 [6]:

- POL0001: Policy error.
- POL0002: Privacy error.

- POL0004: Unlimited notifications not supported.

8.4.4 Operation: endMyWatchersNotificationRequest

Indicates that the presentity does not want further notifications of watcher subscriptions for a specific notification request (identified by the correlator).

8.4.4.1 Input message: endMyWatchersNotificationRequest

Part name	Part type	Optional	Description
correlator	xsd:string	No	Correlator of request to end.

8.4.4.2 Output message: endMyWatchersNotificationResponse

Part name	Part type	Optional	Description
None			

8.4.4.1 Referenced faults

ServiceException from 3GPP TS 29.199-1 [6]:

- SVC0001: Service error.
- SVC0002: Invalid input value.

PolicyException from 3GPP TS 29.199-1 [6]:

- POL0001: Policy error.

8.5 Interface: PresenceSupplierNotification

This client interface is used to notify the presentity of watcher subscriptions.

8.5.1 Void

8.5.2 Operation: notifyError

The notifyError operation is invoked to indicate that the notification for a presentity, or for the whole notification, is being cancelled by the Web Service.

8.5.2.1 Input message: notifyErrorRequest

Part name	Part type	Optional	Description
Correlator	xsd:string	No	Correlator provided in request to set up this notification.
Presentity	xsd:anyURI	Yes	Presentity identifier if the error applies to an individual presentity, or not specified if it applies to the whole notification.
Reason	common:ServiceError	No	Reason notification is being discontinued.

8.5.2.2 Output message: notifyErrorResponse

Part name	Part type	Optional	Description
None			

8.5.2.3 Referenced faults

None.

8.5.3 Void

8.5.4 Operation: notifyMyWatchersRequest

The asynchronous operation is called by the Presence Web Service to inform the client of all watcher subscriptions.

The Watchers part is a structure that includes the identity of watching subscribers for the presentity. The watchers" subscription status and the event that caused the transition to this status are included.

The client can answer open requests (i.e. those with a Watcher Subscription Status of "Pending" or "Waiting") with updateSubscriptionAuthorization().

8.5.4.1 Input message: notifyMyWatchersRequest

Part name	Part type	Optional	Description
correlator	xsd:string	No	Correlator.
Presentity	xsd:anyURI	No	The presentity that the watcher subscription(s) refer to.
Watchers	Watcher [0..unbounded]	Yes	The list of watcher identities that currently have requested access to the presentity's attributes. The subscription status of the watcher is included in the response and the event that caused the transition to the current status.

8.5.4.2 Output message: notifyMyWatchersResponse

Part name	Part type	Optional	Description
None			

8.5.4.3 Referenced faults

None.

8.5.5 Operation: notifyMyWatchersEndRequest

The notifications have completed for this correlator. This message will be delivered when the duration for notifications has expired. This message will not be delivered in the case of an error ending the notifications or deliberate ending of the notifications (using endMyWatchersNotification operation).

8.5.5.1 Input message: notifyMyWatchersEndRequest

Part name	Part type	Optional	Description
correlator	xsd:string	No	Correlator provided in request to set up this notification.

8.5.5.2 Output message: notifyMyWatchersEndResponse

Part name	Part type	Optional	Description
None			

8.5.5.3 Referenced faults

None.

9 Fault definitions

9.1 ServiceException

From 3GPP TS 29.199-1 [6].

9.1.1 SVC0220: No subscription request

Name	Description
Message Id	SVC0220
Text	No subscription request from watcher %1 for attribute %2
Variables	%1 - watcher URI %2 - type of attribute, from clause 7.1

9.1.2 SVC0221: Not a watcher

Name	Description
Message Id	SVC0221
Text	%1 is not a watcher
Variables	%1 - watcher URI

10 Service policies

Name	Type	Description
MaximumNotificationFrequency	common:TimeMetric	Maximum rate of notification delivery (also can be considered minimum time between notifications)
MaximumNotificationDuration	common:TimeMetric	Maximum amount of time a notification may be set up for
DefaultNotificationDuration	common:TimeMetric	Default amount of time a notification will be set up for.
MaximumCount	xsd:int	Maximum number of notifications that may be requested
UnlimitedCountAllowed	xsd:boolean	Allowed to specify unlimited notification count (i.e. either by not specifying the optional Count message part in StartPresenceNotificationRequest or by specifying a value of zero)
GroupSupport	xsd:boolean	Groups may be included with addresses
NestedGroupSupport	xsd:boolean	Are nested groups supported in group definitions
MaximumIdentifiers	xsd:int	Maximum number of allowed URIs provided in a request. A group URI shall be considered as one URI.

Annex A (normative): WSDL of Presence API

The document/literal WSDL representation of this interface specification is compliant to 3GPP TS 29.199-1 [6] and is contained in text files (contained in archive 29199-14-800-doclit.zip) which accompanies the present document.

Annex B (informative): Bibliography

IETF RFC 4660: "Functional Description of Event Notification Filtering". http://www.ietf.org/rfc/rfc4660.txt.

IETF RFC 4480: "RPID: Rich Presence: Extensions to the Presence Information Data Format (PIDF)". http://www.ietf.org/rfc/rfc4480.txt

IETF RFC 4825: "The Extensible Markup Language (XML) Configuration Access Protocol (XCAP)". http://www.ietf.org/rfc/rfc4825.txt.

Repository of information about the Extensible Messaging and Presence Protocol (XMPP), which was contributed by the Jabber Software Foundation (JSF) to the IETF. http://www.xmpp.org/

Annex C (informative): Description of Parlay X Web Services Part 14: Presence for 3GPP2 cdma2000 networks

This annex is intended to define the OSA Parlay X Web Services Stage 3 interface definitions and it provides the complete OSA specifications. It is an extension of OSA Parlay X Web Services specifications capabilities to enable operation in cdma2000 systems environment. They are in alignment with 3GPP2 Stage 1 requirements and Stage 2 architecture defined in:

[1] 3GPP2 X.S0011-D: "cdma2000 Wireless IP Network Standard ", Version 1.1

[2] 3GPP2 S.R0037-0: "IP Network Architecture Model for cdma2000 Spread Spectrum Systems", Version 3.0

[3] 3GPP2 X.S0013-A: "All-IP Core Network Multimedia Domain"

These requirements are expressed as additions to and/or exclusions from the 3GPP Release 8 specification.
The information given here is to be used by developers in 3GPP2 cdma2000 network architecture to interpret the 3GPP OSA specifications.

C.1 General Exceptions

The terms 3GPP and UMTS are not applicable for the cdma2000 family of standards. Nevertheless these terms are used (3GPP TR 21.905) mostly in the broader sense of "3G Wireless System". If not stated otherwise there are no additions or exclusions required.

CAMEL mappings are not applicable for cdma2000 systems.

C.2 Specific Exceptions

C.2.1 Clause 1: Scope

There are no additions or exclusions.

C.2.2 Clause 2: References

There are no additions or exclusions.

C.2.3 Clause 3: Definitions and abbreviations

There are no additions or exclusions.

C.2.4 Clause 4: Detailed service description

There are no additions or exclusions.

C.2.5 Clause 5: Namespaces

There are no additions or exclusions.

C.2.6 Clause 6: Sequence diagrams

There are no additions or exclusions.

C.2.7 Clause 7: XML Schema data type definition

There are no additions or exclusions.

C.2.8 Clause 8: Web Service interface definition

There are no additions or exclusions.

C.2.9 Clause 9: Fault definitions

There are no additions or exclusions.

C.2.10 Clause 10: Service policies

There are no additions or exclusions.

C.2.11 Annex A (normative): WSDL of Presence API

There are no additions or exclusions.

Annex D (informative): Change history

Change history								
Date	**TSG #**	**TSG Doc.**	**CR**	**Rev**	**Subject/Comment**	**Cat**	**Old**	**New**
Dec 2005	CT-30	CP-050583	0007	--	Parameterization for requester of Presence Web service	B	6.3.0	7.0.0
Dec 2005	CT-30	CP-050583	0008	--	Adding communication means" Status in Presence Web service	B	6.3.0	7.0.0
Jun 2006	CT-32	CP-060203	0010	--	Apply Union data type element naming convention	A	7.0.0	7.1.0
Dec 2006	CT-34	CP-060594	0012	--	Corrections to text descriptions to remove ambiguity	A	7.1.0	7.2.0
Dec 2006	CT-34	CP-060606	0013	--	Update the description of "requester" message part to clarify its purpose and usage	F	7.1.0	7.2.0
Mar 2007	CT-35	CP-070045	0015	-	Add OSA Parlay Web Services support for 3GPP2 networks	A	7.2.0	7.3.0
Mar 2007	CT-35	CP-070045	0021	-	Remove inconsistencies of indicating subscription scope	A	7.2.0	7.3.0
Mar 2007	CT-35	CP-070048	0016	--	Add support for Polite Block	B	7.2.0	7.3.0
Mar 2007	CT-35	CP-070048	0017	--	Extend get watcher request and add subscription notification interface	B	7.2.0	7.3.0
Mar 2007	CT-35	CP-070048	0018	--	Add support for array of presentities	C	7.2.0	7.3.0
Mar 2007	CT-35	CP-070048	0019	--	Alignment of presence attributes types with IETF	F	7.2.0	7.3.0
Mar 2007	--	--	--	--	Editorial: Aligned 5 Namespaces	--	7.3.0	7.3.1
Jun 2007	--	--	--	--	Renamed in Introduction Part 18:"Device management" to "Device Capabilities and Configuration". Updated Annex B Bibliography - input on IETF from Julian	--	7.3.1	7.3.2
Sep 2007	CT-37	CP-070640	0023	--	Remove SVC0005 from Service Exception of SubscibePresence	F	7.3.2	7.4.0
Dec 2007	--	--	--	--	Code attachment provided by PTCC.	--	7.4.0	7.4.1
May 2008	CT-40	CP-080255	0024	--	Allow an updated Watcher structure to be returned by the getOpenSubscriptions and getMyWatchers operations instead of the SubscriptionRequest structure.	C	7.4.1	8.0.0
May 2008	CT-40	CP-080255	0025	--	Introduce the updateAuthorizationRule operation with parts to match the more complex structure of authorization / permissions supported by SIP / IMS networks. Also add a deleteAuthorizationRule operat	C	7.4.1	8.0.0
May 2008	CT-40	CP-080255	0026	--	Add the new methods startMyWatchersNotification and endMyWatchersNotification and rename the existing methods startSubscriptionNotification and EndNotification as startOpenSubscriptionNotification and	C	7.4.1	8.0.0
May 2008	CT-40	CP-080255	0027	--	Deletion of Methods in Parlay-X Presence	C	7.4.1	8.0.0
May 2008	CT-40	CP-080255	0028	--	Update the call flow section for the Presence APIs to reflect the inclusion of support for SIP SIMPLE	C	7.4.1	8.0.0
May 2008	CT-40	CP-080255	0029	--	Provide the possibility to use a one stage startPresenceNotification or getUserPresence operation instead of the two stage subscribePresence / startPresenceNotification and subscribePresence / getUser	C	7.4.1	8.0.0
May 2008	CT-40	CP-080255	0030	--	Allow the two stage notifySubscription / statusChanged operations to be avoided by the use of a newly defined one stage statusNotified operation that also supports notifications for multiple presentit	C	7.4.1	8.0.0
May 2008	CT-40	CP-080255	0031	--	Add the new methods notifyMyWatchers and notifyMyWatchersEnd and rename the existing method notifyEnd as notifyOpenSubscriptionEnd	C	7.4.1	8.0.0

History

Document history		
V8.0.0	January 2009	Publication